

Computer-assisted proof candidate for the all- n odd-cycle assertion in Erdős Problem 883

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Version 3, 4 October 2026 UTC

Review edition prepared from the v3 source with documented editorial clarifications

Abstract

We give a proof candidate for the first question of Erdős 883 for every finite n . The proof uses odd totient-profile smoothing, small-prime signature ordering and an ordered-Hall construction. An explicit analytic argument handles $n \geq 2,000,000$. Exact interval certificates cover every smaller n , for every permitted subset A . No regularity lemma or unspecified starting threshold remains in the proposed argument.

The proposed theorem

For every integer $n \geq 0$ and every $A \subseteq \{1, \dots, n\}$ satisfying

$$|A| > \left\lfloor \frac{n}{2} \right\rfloor + \left\lfloor \frac{n}{3} \right\rfloor - \left\lfloor \frac{n}{6} \right\rfloor,$$

the graph on A with distinct vertices u, v adjacent exactly when $\gcd(u, v) = 1$ contains a simple cycle of every odd length L with $3 \leq L \leq n/3 + 1$. Equivalently, it contains $C_{2\ell+1}$ for every integer $1 \leq \ell \leq \lfloor n/6 \rfloor$.

Status and attribution

This is an AI-assisted proof candidate whose mathematical arguments and exact certificates have received independent AI-assisted adversarial checks. It has not received external peer review or Lean/formal verification. No first-solution, novelty or priority claim is made.

The proof explicitly adapts Donald Della Pietra’s earlier missing-even smoothing, totient profiles, prime signatures and ordered-Hall method. His pinned 27 July 2026 manuscript states a sufficiently-large- n theorem [3]. The claimed addition here is the exact finite completion and explicit threshold. The separate complete-tripartite question is not a new claim.

AI assistance disclosure. This manuscript, its computational certificates and its review preparation were developed with substantial AI assistance. The listed author name is Yicheng Pan (潘奕成). AI-assisted checks do not substitute for independent external mathematical review.

Organization and conventions

Part I proves an exact interval criterion covering all subsets A . Part II provides adaptive arithmetic bounds and a profile-only certificate. Part III proves the exact finite-prefix totient estimates. Part IV proves the theorem for every $n \geq 2,000,000$. Part V documents the complete exact finite bridge. Section numbers mentioned within a part are local to that part.

All decimal constants in proof inequalities denote exact terminating rationals. Displayed decimal approximations in checker output are diagnostic; no floating-point comparison determines a certificate

pass. The two decimal expansions of rational error bounds in Part III are explicitly marked as approximations in version 3 and this review edition. The exact rational bounds are authoritative. Appendix A records the limited editorial changes made for this edition.

The only external theorem

Ahlsvede–Khachatrian (1996), Theorem 2(i), specialized to odd integers, bounds an intersecting odd family by $\lfloor n/3 \rfloor - \lfloor n/6 \rfloor$ [1]. We use it only for $n \geq 6$ and a retained set of at least two odd vertices. If that set contains 1, it already has a coprime pair; otherwise absence of a coprime pair makes it intersecting under their convention even when equal pairs are included. This avoids any singleton-1 convention at small n . Theorem 2(i) is on printed page 149 of the primary paper.

Part I Exact finite interval criterion

Notation and external input

Let

$$\begin{aligned} f(n) &= \lfloor n/2 \rfloor + \lfloor n/3 \rfloor - \lfloor n/6 \rfloor, & m(n) &= \lfloor n/6 \rfloor, \\ H(n) &= \lceil n/2 \rceil, & M(n) &= \lfloor n/3 \rfloor - \lfloor n/6 \rfloor + 1. \end{aligned}$$

Let O_U be the odd integers in $[U]$, in any fixed total order (the implementation uses decreasing $\phi(v)/v$, breaking ties by increasing v). Write P_k for its first k members.

The sole non-elementary arithmetic input is Ahlsvede–Khachatrian (1996), Theorem 2, specialized to $s = 2$: a set of pairwise noncoprime odd integers in $[n]$ has size at most $\lfloor n/3 \rfloor - \lfloor n/6 \rfloor$. Thus any $M(n)$ distinct odds contain two coprime elements. The primary source and bibliographic identity are given in [1].

Choose any ordered finite list of odd primes $p_1, \dots, p_t$. The s -signature of v is $(1_{p_i|v})_{1 \leq i \leq s}$. Set $h_0 = 0$ and $h_s = 2^s + 1$ for $s \geq 1$. For integers $6 \leq L \leq U$, define:

$$\begin{aligned} e_s(k; L, U) &= \min_{\substack{u, v \in P_k, u \neq v \\ \text{equal } s\text{-signature}}} \#\{\text{even } w \leq L : \gcd(w, u) = \gcd(w, v) = 1\}; \\ r_s(k; L, U) &= \text{the same minimum with all positive } w \leq L. \end{aligned}$$

If there is no such pair, set the corresponding minimum to $+\infty$. Finite sentinels exceeding every tested threshold may be used; the implementation uses $\lfloor U/2 \rfloor + 1$ and $U + 1$.

Exact interval criterion

Suppose for every integer

$$0 \leq b \leq H(U) - M(U), \quad 1 \leq j \leq m(U),$$

there exist an $s \in \{0, \dots, t\}$ and $k \in \{0, \dots, H(U)\}$ such that

$$2 \max(0, H(U) - k - b) + h_s < j, \tag{1}$$

and at least one of

$$e_s(k; L, U) \geq b + j, \tag{2E}$$

$$r_s(k; L, U) \geq U - f(U) + m(U) + j. \tag{2R}$$

Then for every integer $n \in [L, U]$ and every $A \subseteq [n]$ with $|A| > f(n)$, $G(A)$ contains every odd cycle $C_{2\ell+1}$, $1 \leq \ell \leq \lfloor n/6 \rfloor$.

Proof

Fix such n, A and put $b = \#(\text{even integers in } [n] \text{ outside } A)$. There are at least $M(n) + b$ selected odds. Retain the first $M(n)$ of these in the fixed O_U order, calling the resulting set Y_0 . For every k ,

$$|Y_0 \setminus P_k| \leq \max(0, H(U) - k - b). \quad (3)$$

Indeed, if a retained odd lies after position k , every discarded selected odd lies after k as well; there are at least b discarded odds. Counting inside O_U proves (3). If no retained odd lies after k the assertion is immediate.

By the stated Ahlswede–Khachatrian theorem, choose coprime distinct a, a' in Y_0 . Since $M(n) \geq m(n) + 1$, for each $\ell \leq m(n)$ choose a skeleton of $\ell + 1$ members of Y_0 containing a, a' . Sort this skeleton lexicographically by its full t -signatures. Move a to the first position and a' to the last, without changing the relative order of the other vertices. Denote the result $o_0 = a, \dots, o_\ell = a'$.

This one order has, simultaneously for every $s \geq 1$, at most h_s gaps joining different s -signatures: before endpoint moves, every prefix-signature occupies one block and there are at most $2^s - 1$ transitions; deleting a vertex cannot increase the number of transitions (the 0/1 inequality metric obeys the triangle inequality), and inserting each prescribed endpoint at an end increases it by at most one. For $s = 0$ there are no transitions. At most twice the number in (3) of the gaps touch an odd outside P_k .

Let $R = A$ minus the skeleton and let

$$D_i = \{w \in R : \gcd(w, o_{i-1}) = \gcd(w, o_i) = 1\}.$$

For each rank $j \leq \ell$ take s, k from the criterion. Every gap except at most $2 \max(0, H(U) - k - b) + h_s$ has both endpoints in P_k and equal s -signature.

If (2E) holds, that gap has at least $b + j$ common even neighbors in $[L]$. At most b are missing from A , and no even vertex is in the odd skeleton, so $|D_i| \geq j$.

If (2R) holds, it has at least $U - f(U) + m(U) + j$ common neighbors in $[L]$. Removing vertices outside A and in the skeleton costs at most

$$(n - |A|) + (\ell + 1) \leq n - f(n) + \ell \leq U - f(U) + m(U),$$

so again $|D_i| \geq j$. The last inequality follows from monotonicity of $n - f(n) + m(n)$; writing $n = 6q + r$, it equals $3q + (0, 1, 1, 1, 1, 2)_r$. Also $b \leq H(n) - M(n) \leq H(U) - M(U)$, since $H(n) - M(n) = 2q + (-1, 0, 0, 0, 0, 1)_r$ is nondecreasing.

Consequently fewer than j lists D_i have size less than j . Sort the ℓ lists by increasing cardinality. The j th list has size at least j , so greedy selection supplies distinct representatives w_i . They all lie outside the skeleton. Hence

$$o_0, w_1, o_1, \dots, w_\ell, o_\ell, o_0$$

is a simple coprime cycle of length $2\ell + 1$, closing via $\gcd(a, a') = 1$. Different ranks may use different s, k or (2E)/(2R), because each bounds the same actual resource lists D_i . This proves the criterion.

Why the implementation checks only one k for each s, b, j

If $h_s \geq j$ there is no feasible k . Otherwise the smallest feasible k is

$$k = H(U) - b - \left\lfloor \frac{j - h_s - 1}{2} \right\rfloor.$$

Both common-neighbor minima are nonincreasing in k , so this k is optimal for satisfying (2E)/(2R). In the ranges tested $k \geq 2$. All sorting uses exact rational $\phi(v)/v$; all common-neighbor counts use exact integer gcd and bitsets; there is no floating-point arithmetic in the certificate conditions.

Part II Adaptive arithmetic and efficient certificates

Notation

$\rho(v) = \phi(v)/v$. Throughout this part, endpoint pairs u, v are odd integers. The first s odd primes are $p_1 = 3, \dots, p_s$; $p = p_{s+1}$. For $s = 0$, set $p_0 = 2$. The 0-signature is empty, $p = p_1 = 3$, and the tail primes $r > p_0$ are exactly the odd primes; all endpoint pairs remain odd. Let U be an integer upper bound on every skeleton vertex and let $L \leq U$. Let

$$\begin{aligned} H &= \lceil U/2 \rceil, & q_L &= \lfloor L/2 \rfloor, & m &= \lfloor U/6 \rfloor, \\ f &= \lfloor U/2 \rfloor + \lfloor U/3 \rfloor - \lfloor U/6 \rfloor, & M &= f - \lfloor U/2 \rfloor + 1. \end{aligned}$$

For $0 < z \leq 1$ define the strict finite profile count

$$R_U(z) = \#\{\text{odd } v \leq U : \rho(v) < z\}.$$

Strict inequality ensures that every remaining vertex has profile $\geq z$. Write $h_0 = 0$ and $h_s = 2^s + 1$ for $s \geq 1$.

1. Exact adaptive tail factor

Let a be the greatest nonnegative integer with $p^a \leq U$ and put

$$\eta_s(U) = \left(\frac{p-1}{p} \right)^a.$$

Every $v \leq U$ has at most a distinct prime divisors exceeding p_s . Each corresponding totient factor is at least $(p-1)/p$. Therefore

$$\prod_{\substack{r|v, r > p_s \\ r \text{ prime}}} \left(1 - \frac{1}{r} \right) \geq \eta_s(U).$$

All ingredients are integers or rational numbers; logarithms are unnecessary for computing a .

If odd integers $u, v \leq U$ share their first s prime-divisibility signatures, every prime factor of v absent from u exceeds p_s . Hence

$$\begin{aligned} \rho(\text{lcm}(u, v)) &= \rho(u) \prod_{\substack{r|v \\ r \nmid u}} \left(1 - \frac{1}{r} \right) \\ &\geq \rho(u) \eta_s(U). \end{aligned}$$

Interchanging u, v proves the slightly stronger bound

$$\rho(\text{lcm}(u, v)) \geq \max(\rho(u), \rho(v))\eta_s(U).$$

In particular it is $\geq z\eta_s(U)$ when both profiles are $\geq z$.

One may sharpen $\eta_s(U)$ without changing the proof. Let r be maximal such that the product of the first r primes after p_s is $\leq U$. No integer $\leq U$ has more than r distinct tail prime factors. Among any r such primes the smallest r give the smallest totient-factor product. Thus their factor product is a valid sharper rational $\eta_s(U)$.

2. Exact common-neighbor discrepancy

For odd squarefree D with $w = \omega(D) \geq 1$ and integer $X \geq 0$,

$$\begin{aligned} C_D(X) &= \#\{1 \leq a \leq X : \gcd(a, D) = 1\} \\ &= \sum_{d|D} \mu(d) \lfloor X/d \rfloor. \end{aligned}$$

Comparing with $X\phi(D)/D$, only the positive- μ terms can contribute negatively. There are 2^{w-1} such divisors, and $d = 1$ has zero fractional part. Thus

$$C_D(X) \geq X\rho(D) - (2^{w-1} - 1).$$

For $D = 1$ the equality $C_1(X) = X$ holds.

Define $W(U)$ as the greatest w such that the product of the first w odd primes is $\leq U^2$, and $\Delta(U) = 2^{W(U)-1} - 1$ (take zero if $W(U) = 0$). For $D = \text{rad}(uv)$, $D \leq uv \leq U^2$, so the previous inequality gives

$$\begin{aligned} \text{common even neighbors in } [L] &\geq q_L\rho(D) - \Delta(U), \\ \text{common whole-pool neighbors in } [L] &\geq L\rho(D) - \Delta(U). \end{aligned}$$

This does not require D to divide L and remains valid if an endpoint is 1.

3. Exact finite criterion using only a univariate profile

For every required pair $0 \leq b \leq H - M$ and $1 \leq j \leq m$, suppose one can choose a signature prefix s and rational z with

$$2 \max(0, R_U(z) - b) + h_s < j, \tag{A}$$

and either

$$q_L z \eta_s(U) - \Delta(U) \geq b + j, \tag{B_E}$$

or

$$L z \eta_s(U) - \Delta(U) \geq U - f + m + j. \tag{B_R}$$

Then the all-subsets conclusion of the finite interval criterion holds for every $n \in [L, U]$.

Proof: retain the best $M(n)$ selected odds as before. Among the $R_U(z)$ low profile odds, at most $\max(0, R_U(z) - b)$ can remain in that retained set, because at least b selected odds worse than any retained low-profile odd were discarded. Order the skeleton by a single sufficiently long prime signature and move its coprime endpoints to the ends. For any selected prefix s , all but the number in (A) of its gaps have equal signatures and both endpoint profiles $\geq z$. Sections 1 and 2 give at least j resources for each such gap. The ordered-Hall proof applies unchanged.

Different ranks may select different prefix lengths. A single ordering by the longest selected prefix has all shorter prefixes contiguous before moving endpoints, so there is no consistency problem.

4. One fully uniform tail bound with only $O(\sqrt{U})$ transitions

For every integer $U > 1024$ choose the least s with $U \leq 4^s$. Thus $s \geq 6$ and $2^s < 2\sqrt{U}$ (or the weaker $\leq$ suffices). A valid tail factor for every $v \leq U$ is at least $9/10$. Consequently

$$h_s \leq 2\sqrt{U} + 1,$$

$$\rho(\text{lcm}(u, v)) \geq \frac{9}{10} \max(\rho(u), \rho(v))$$

whenever the two signatures agree.

Proof for $s \geq 511$: the next odd prime $p \geq 2s + 3 > 1024$. If a is the number of distinct tail primes in v , then $p^a \leq U \leq 4^s$, giving

$$\frac{a}{p} \leq \frac{s \log 4}{p \log p} < \frac{\log 2}{\log(2s + 3)} < \frac{1}{10}.$$

Bernoulli's inequality gives $(1 - 1/p)^a \geq 1 - a/p > 9/10$. For $6 \leq s \leq 510$, exact rational arithmetic checks the sharper tail factor from Section 1 with upper bound 4^s . Every case is $\geq 9/10$. The smallest factor is $396/437$ at $s = 6$, from the two tail primes 19 and 23. This finite parameter calculation is included in `verify_adaptive_tail_lemma.py`; it is not enumeration over n or over subsets A . The standalone verifier reported about 0.04 CPU seconds.

A useful uniform discrepancy bound is also elementary. For odd squarefree D ,

$$\frac{(2^{\omega(D)})^4}{D} = \prod_{p|D} \frac{16}{p} \leq \frac{16^5}{3 \cdot 5 \cdot 7 \cdot 11 \cdot 13} < 81,$$

since factors from primes ≥ 17 are ≤ 1 . Hence $2^{\omega(D)} < 3D^{1/4}$. For $D \leq U^2$, the actual discrepancy from Section 2 is strictly less than $(3/2)\sqrt{U}$. Thus $(B_E)/(B_R)$ may use loss $(3/2)\sqrt{U}$ and factor $9/10$, without computing $W(U)$.

This proves a universal numerical reduction for all $U > 1024$:

$$2 \max(0, R_U(z) - b) + 2^s + 1 < j,$$

and either

$$\frac{9}{10} q_L z - \frac{3}{2} \sqrt{U} \geq b + j,$$

or

$$\frac{9}{10} L z - \frac{3}{2} \sqrt{U} \geq U - f + m + j$$

are sufficient. The exact integer $2^s + 1$ is preferable to its displayed $2\sqrt{U} + 1$ upper bound. Irrational square roots may be rounded upward in the subtracted error if a rational-only certificate is desired.

7. Stronger density bound and independent audit of the fast finite test

Always

$$\rho(\text{lcm}(u, v)) = \frac{\rho(u)\rho(v)}{\rho(\text{gcd}(u, v))} \geq \rho(u)\rho(v).$$

Consequently, for both profiles $\geq z$ and equal s -signatures, the valid lower bound is

$$g_s(z) = \max(z^2, \eta_s(U)z).$$

This may replace $z\eta_s(U)$ everywhere in Section 3, and substantially improves shorter prefixes. In particular, the z^2 term needs no signatures at all and can be used at $s = 0$, where $h_0 = 0$.

For each odd $v \leq U$ define integer lower-degree functions

$$\begin{aligned} d_E(v) &= \lfloor qLg_s(\rho(v)) \rfloor - \Delta(U), \\ d_R(v) &= \lfloor Lg_s(\rho(v)) \rfloor - \Delta(U). \end{aligned}$$

Both are nondecreasing functions of $\rho(v)$. If two endpoints each have $d_E(v) \geq K$ and share their s -signatures, their actual common-even-neighbor count is $\geq K$. The analogous assertion holds for d_R . Thus define

$$\begin{aligned} R_E(K) &= \#\{\text{odd } v \leq U : d_E(v) < K\}, \\ R_R(K) &= \#\{\text{odd } v \leq U : d_R(v) < K\}. \end{aligned}$$

For fixed s, j with $h_s < j$, put $a = \lfloor (j - h_s - 1)/2 \rfloor$, $D = U - f + m$. The needed certificate for a given b is exactly the sufficient disjunction

$$R_E(b + j) \leq b + a \quad \text{OR} \quad R_R(D + j) \leq b + a.$$

The whole-resource alternative covers all $b \geq R_R(D + j) - a$. For smaller b , put $z = b + j$ and $Z(z) = R_E(z) - z$. Checking all b reduces to

$$\max_{j \leq z \leq \min(j + b_{\max}, j + R_R(D + j) - a - 1)} Z(z) \leq a - j,$$

with an empty range automatically passing.

Both range endpoints are nondecreasing in j : $R_R(D + j)$ is nondecreasing, and $j - a$ is nondecreasing because a increases only every other step. Therefore the monotone-deque range maximum in `adaptive_density_probe.py` is correct. Finding one s that passes this entire b -range for each j is a stronger sufficient quantifier than allowing s to depend on b , and is valid.

The supplied implementations use exact integer comparisons for both rational densities. Histogram index $d + 1$ and prefix sums represent strict degree thresholds. Clamping negative degrees to -1 is harmless because all queried thresholds are positive. Prime-list cutoff completeness is asserted. The C++ certificate uses signed 128-bit intermediate products, with the input restricted to $U \leq 2,000,000$. These bounds are far below overflow.

Part III Guarded finite prefix totient estimates

Main results

Write

$$G_n(t) = \frac{2}{n} \#\{v \leq n : v \text{ odd and } \phi(v)/v < t\};$$

normalization is by $n/2$, not $\lceil n/2 \rceil$, so the bounds plug directly into the graph estimates.

For *every* integer $n \geq 1,000,000$ and *every* real $t \in [1/3, 3/4]$,

$$G_n(t) < \frac{9}{10}t - \frac{1}{6} - \frac{83}{10000}. \tag{H}$$

This interval includes the needed endpoint $20/27$ and guards it through .75.

For *every* integer $n \geq 2,000,000$ and *every* $\beta \in [1/10, 201/100]$,

$$G_n \left(\sqrt{\frac{\beta + 3/100}{3}} \right) < \frac{\beta}{3} - \frac{33}{2500}. \quad (\text{L})$$

The remaining margin is .0132. In particular, before finite-prefix error, the proposed .02 low-rank margin succeeds, with room to spare.

Both statements use only 16 states for the primes 3, 5, 7, 11 and elementary first-/sixth-moment estimates. No Mertens theorem or Rosser–Schoenfeld bound is required. The exact checker uses < 1 CPU-second and enumerates primes only through 2000. It does not enumerate graph subsets A .

Proof

1. Constants, all checked with exact fractions

Let $\lambda_p = -\log(1 - 1/p)$, and let $P = \{3, 5, 7, 11\}$.

$$\begin{aligned} \delta_{\text{actual}} &= \sum_{p>11} \frac{\lambda_p}{p} < D := \frac{47}{2000} = .0235, \\ C_6 &= \prod_{\text{odd } p} \left(1 + \frac{(p/(p-1))^6 - 1}{p} \right) < 11. \end{aligned}$$

For each odd prime $p \leq 2000$, the first 12 terms of

$$\lambda_p = \sum_{h \geq 1} \frac{1}{hp^h}$$

give a lower bound, with remainder at most $1/[13p^{12}(p-1)]$. Sum the resulting upper bounds for λ_p/p and add $1/2000$ for all omitted primes: $\lambda_p/p \leq 1/[p(p-1)]$, whose tail over all integers $p > 2000$ telescopes to $1/2000$. Subtracting the four small-prime lower bounds gives the displayed upper bound for δ_{actual} .

For C_6 , multiply its Euler factors through $p \leq 200$. For $p > 200$ the binomial expansion gives $(p/(p-1))^6 - 1 \leq 7/(p-1)$. Thus the omitted logarithmic tail is $\leq 7/200$ and the product tail is $\leq \exp(7/200) \leq 200/193$.

Every finite sum and product is evaluated as an exact rational in the bundled checker. Verified upper values:

$$\delta_{\text{actual}} < .023456772181318, \quad C_6 < 10.994755089901.$$

2. Uniform sixth moment, with no error term

For odd squarefree d , put

$$c(d) = \prod_{p|d} \left(\left(\frac{p}{p-1} \right)^6 - 1 \right),$$

so

$$\left(\frac{v}{\phi(v)} \right)^6 = \sum_{d|v} c(d),$$

with nonnegative coefficients. The number of odd multiples of d in $[n]$ is at most n/d . Therefore

$$\frac{2}{n} \sum_{\substack{v \leq n \\ v \text{ odd}}} \left(\frac{v}{\phi(v)} \right)^6 \leq 2 \sum_d \frac{c(d)}{d} = 2C_6 < 22,$$

and Markov gives $G_n(t) < 22t^6$, uniformly for *every* n .

3. The 16-state bound and its finite-prefix discrepancy

For $S \subseteq P$ put

$$\begin{aligned} a_S &= \prod_{p \in S} \left(1 - \frac{1}{p} \right), \\ q_S &= \prod_{p \in S} \frac{1}{p} \prod_{p \in P \setminus S} \left(1 - \frac{1}{p} \right), \\ e_S &= 2^{3-|S|}. \end{aligned}$$

Then $\sum q_S = 1$ and $\sum e_S = 81/2$. For any odd d , its odd-multiple count $K_d(n)$ is $\lfloor n/(2d) + 1/2 \rfloor$, so

$$\left| K_d(n) - \frac{n}{2d} \right| \leq \frac{1}{2}.$$

By four-prime inclusion-exclusion, a signature count K_S satisfies

$$\left| K_S - q_S \frac{n}{2} \right| \leq e_S.$$

For every prime $p > 11$, the count $K_{S,p}$ with that signature and $p \mid v$ satisfies

$$\left| K_{S,p} - q_S \frac{n}{2p} \right| \leq e_S.$$

Consequently, for $T(v) = \sum_{p > 11, p \mid v} \lambda_p$,

$$\begin{aligned} \sum_{\substack{v \leq n, v \text{ odd} \\ \text{signature } S}} T(v) &\leq q_S \frac{n}{2} D + e_S L(n), \\ L(n) &= \sum_{11 < p \leq n} \lambda_p. \end{aligned}$$

For $n \geq 13$ the elementary odd-integer comparison gives the sharp-enough bound

$$\begin{aligned} L(n) &\leq \sum_{j=6}^{\lfloor (n-1)/2 \rfloor} \frac{1}{2j} \\ &\leq \frac{1}{2} \log \left(\frac{\lfloor (n-1)/2 \rfloor}{5} \right) \\ &\leq \frac{1}{2} \log(n/10). \end{aligned} \tag{1}$$

No additive $+1$ is necessary. Here $\lambda_p \leq 1/(p-1)$, and the harmonic-sum bound follows by integrating $1/x$ from 5 to $\lfloor (n-1)/2 \rfloor$.

If $a_S \leq t$, use the signature count. Otherwise, Markov bounds that signature's contribution to $G_n(t)$ by

$$\min\left(q_S + \frac{2e_S}{n}, \frac{q_S D + 2e_S L(n)/n}{\log(a_S/t)}\right).$$

Use the rational lower bound $r_S(t) = 2(a_S - t)/(a_S + t) \leq \log(a_S/t)$, and set

$$B_*(t) = \sum_S q_S u_S(t),$$

$$u_S(t) = \begin{cases} 1, & \text{if } a_S \leq t, \\ \min\left(1, \frac{D(a_S + t)}{2(a_S - t)}\right), & \text{if } a_S > t. \end{cases}$$

This function is nondecreasing.

For the error comparison, if $r_S < D$ the limiting summand is q_S and the signature-count error is $\leq 2e_S/n$. If $r_S \geq D$, the extra Markov term is at most $2e_S L(n)/(Dn)$. For the n at issue L 's upper bound can be taken $\geq D$. Summing proves

$$G_n(t) \leq B_*(t) + E(n),$$

$$E(n) = \frac{81}{2D} \frac{\log(n/10)}{n} \tag{2}$$

$$= \frac{81000}{47} \frac{\log(n/10)}{n}.$$

This comparison *does not* require an extra '+1' in $E(n)$. Use the moment bound with $B(t) = \min(22t^6, B_*(t))$ to get $G_n(t) \leq B(t) + E(n)$.

4. Guarded high-branch certificate

Divide $[1/3, 3/4]$ into 1000 cells of width $1/2400$. On every cell $[l, r]$, the exact-rational checker verifies

$$\frac{9}{10}l - \frac{1}{6} - B(r) > \frac{29}{1000}.$$

The minimum actual cell slack is approximately .02905399178577378. By monotonicity, $B(t) < .9t - 1/6 - .029$ everywhere on the interval.

For $n \geq 10^6$ the right side of (2) decreases with n . The exact rational check $\sum_{k=0}^{20} 12^k/k! > 10^5$ proves $\log(10^5) < 12$. Thus

$$E(n) \leq \frac{972000}{47000000},$$

$$.029 - E(n) > .0083,$$

The fraction $972000/47000000$ has diagnostic approximation 0.02068085106382979 . The exact rational bound above proves (H).

5. Low-rank endpoint certificate

Divide $[1/10, 201/100]$ into 1910 beta-cells $[l, r]$ of width $1/1000$. For each right endpoint, bound $\sqrt{(r + .03)/3}$ upward by a rational on the 10^{-9} grid, using integer isqrt and an exact squared comparison. The checker verifies

$$\frac{l}{3} - B\left(\text{upper_sqrt}\left(\frac{r + .03}{3}\right)\right) > \frac{119}{5000} = .0238.$$

The minimum actual cell slack is approximately .02389295162749755. By monotonicity, throughout the interval,

$$B(\sqrt{(\beta + .03)/3}) < \beta/3 - .0238.$$

For $n \geq 2 \cdot 10^6$, the exact check $\sum_{k=0}^{20} (49/4)^k / k! > 2 \cdot 10^5$ proves $\log(2 \cdot 10^5) < 49/4$. Therefore

$$\begin{aligned} E(n) &\leq \frac{3969000}{376000000}, \\ .0238 - E(n) &> .0132, \end{aligned}$$

The fraction $3969000/376000000$ has diagnostic approximation 0.010555851063829787 . The exact rational bound above proves (L).

Part IV Explicit analytic proof above the threshold

Let $\rho(v) = \phi(v)/v$, $H = \lceil n/2 \rceil$,

$$R_n(t) = \#\{\text{odd } v \leq n : \rho(v) < t\}, \quad G_n(t) = \frac{2}{n} R_n(t).$$

The supplied CDF certificate establishes exactly:

- (I) $G_n(t) < 22t^6$ for every n and $t > 0$;
- (II) $G_n(t) < 0.9t - 1/6 - 0.0083$ for $n \geq 1,000,000$, $1/3 \leq t \leq 3/4$;
- (III) $G_n(\sqrt{(\beta + 0.03)/3}) < \beta/3 - 0.0132$ for $n \geq 2,000,000$ and $0.1 \leq \beta \leq 2.01$.

Input (I) follows from $C_6 < 11$, expansion into nonnegative divisor weights, and the upper bound n/d for the number of odd multiples of odd d . In particular it has no additive finite- n error. Inputs (II),(III) already include every finite-prefix error; none should be added again below.

For every $n > 1024$ choose the least s with $n \leq 4^s$ and use the first s odd prime divisibility coordinates. The adaptive-tail certificate establishes that a same-signature pair with both profiles $\geq t$ has common-neighbor density $\geq 0.9t$. Every pair, without a signature condition, has density $\geq t^2$. Inclusion-exclusion gives common-neighbor loss $< (3/2)\sqrt{n}$. Moving the two coprime skeleton endpoints to the ends leaves at most $h = 2^s + 1 < 2\sqrt{n} + 1$ cross-signature gaps. These are the only graph-arithmetic lemmas used below.

1. Exact integer accounting and uniform numerical bounds

Fix $n \geq N = 2,000,000$. Write $n = 6m + r$, $0 \leq r \leq 5$. Let

$$\begin{aligned} q &= \lfloor n/2 \rfloor, & f &= \lfloor n/2 \rfloor + \lfloor n/3 \rfloor - m, \\ M &= f - q + 1, & b &= \#\text{missing even integers}, \\ \beta &= b/m, & x &= j/m \quad \text{for } 1 \leq j \leq m, \\ D &= n - f + m = 3m + (0, 1, 1, 1, 1, 2)_r. \end{aligned}$$

Feasibility gives $0 \leq b \leq H - M$ and $\beta \leq 2 + 1/m$. Also

$$q \geq 3m, \quad H \leq 3m + 3, \quad \frac{n}{2m} < 3.000008, \quad \frac{22H}{m} < 66.001.$$

Let $\delta = (3/2)\sqrt{n}$, and $\epsilon = (\delta + 2)/m$. The proven discrepancy bound for every odd endpoint pair is strictly below δ . Choose the least s with $n \leq 4^s$; its transition budget $h = 2^s + 1$ satisfies $h < 2\sqrt{n} + 1$, and the adaptive tail factor is at least $9/10$.

Uniformly throughout $n \geq N$,

$$\epsilon < 0.0065, \quad h/m < 0.0086. \quad (1)$$

To check these without relying on unrecorded decimals, $m \geq (n - 5)/6$, and the upper functions $(9\sqrt{n} + 12)/(n - 5)$, $(12\sqrt{n} + 6)/(n - 5)$ decrease for $n > 5$. At N use $\sqrt{N} < 1415$; cross multiplication gives both bounds in (1).

For $t \leq 1$, let $G_n(t) = (2/n)R_n(t)$. The integer 1 is never counted in $R_n(t)$, so $0 \leq G_n(t) \leq 1$. Since $n/(2m) < 3.000008$,

$$\frac{R_n(t)}{m} < 3G_n(t) + 0.000008. \quad (2)$$

For the direct moment estimate we may use the slightly weaker convenient bound $R_n(t)/m < 66.001t^6$.

2. Why the chosen thresholds supply sufficient resources

For a shared- s -signature pair with both endpoint profiles $\geq t$, the common neighbor counts are at least

$$\begin{aligned} \text{even pool: } & q\frac{9}{10}t - \delta, \\ \text{whole pool: } & n\frac{9}{10}t - \delta. \end{aligned}$$

For an arbitrary pair, with no signature requirement, they are at least $qt^2 - \delta$ and $nt^2 - \delta$, respectively.

For every rank j the even-pool target is $b + j$; the whole-pool target is $D + j$. These account for all missing evens, or all missing vertices and the entire odd skeleton, respectively.

Retain the best M selected odds. At most $\max(0, R_n(t) - b)$ retained odds have profile $< t$. In a skeleton ordered by the full s -signature and with the two coprime endpoints moved to its ends, at most

$$2\max(0, R_n(t) - b) + h$$

successive gaps are exceptional. If the last expression is $< j$, and one of the common-neighbor targets holds, the rank- j Hall condition follows. For an unstructured, product-density bound one uses $h = 0$, with the same fixed skeleton order. Hence the following rankwise cases are compatible.

3. Large ranks: $x \geq 0.02$

Choose

$$t = \min\left(\frac{\beta + x + \epsilon}{2.7}, \frac{3 + x + \epsilon}{5.4}\right). \quad (4)$$

This t is positive and

$$t < 4.0065/5.4 < 0.743.$$

If the first branch in (4) is minimal, the even lower bound is at least

$$2.7mt - \delta = b + j + 2.$$

If the second is minimal, the whole lower bound is at least

$$5.4mt - \delta = 3m + j + 2 \geq D + j.$$

Thus a required resource target always holds.

3a. The high-profile branch $t \geq 10/27$

The finite CDF lemma gives

$$G_n(t) < 0.9t - 1/6 - 0.0083$$

because $1/3 \leq t < 0.743 < 3/4$ and $n \geq 2,000,000$. Using (2) and $t \leq (\beta + x + \epsilon)/2.7$,

$$\begin{aligned} \frac{R_n(t)}{m} &< 2.7t - 1/2 - 0.0249 + 0.000008 \\ &\leq \beta + x + \epsilon - 0.5249 + 0.000008 \\ &\leq \beta + x/2 - 0.0249 + \epsilon + 0.000008, \end{aligned}$$

since $x \leq 1$. Moreover

$$\begin{aligned} \epsilon + \frac{h}{2m} + 0.000008 &< 0.0065 + 0.0043 + 0.000008 \\ &= 0.010808 < 0.0249. \end{aligned}$$

As $x > h/m$, this proves

$$2 \max\left(0, \frac{R_n(t)}{m} - \beta\right) + \frac{h}{m} < x.$$

The rank- j condition follows with strict inequality.

3b. The low-profile branch $t < 10/27$

The second branch of (4) is $> 5/9$, so the first branch must be minimal. Set $y = \beta + x + \epsilon = 2.7t$; then $0.02 \leq y < 1$. From the direct finite moment bound and $22H/m < 66.001$,

$$\frac{R_n(t)}{m} < 66.001(y/2.7)^6 < 0.171y^6. \quad (5)$$

The polynomial $F(y) = 0.342y^6 - y + 0.0151$ is convex on $[0.02, 1]$, and both endpoint values are strictly negative. Thus

$$0.342y^6 + 0.0086 < y - 0.0065 \leq y - \epsilon = \beta + x \leq 2\beta + x.$$

Together with $x \geq 0.02 > h/m$, this and (5) imply the same strict rank- j condition as in 3a. The direct finite moment bound here has no additive error term.

4. Small ranks with many missing evens: $x \leq 0.02$, $\beta \geq 0.1$

Set $\beta_0 = \min(\beta, 2)$ and

$$t = \sqrt{\frac{\beta_0 + 0.03}{3}}.$$

The clipping handles $n = 6m + 5$, when b can equal $2m + 1$; then

$$0 \leq m(\beta - \beta_0) \leq 1.$$

Using the unstructured product bound, the common even-neighbor count is at least

$$qt^2 - \delta \geq m(\beta_0 + 0.03) - \delta.$$

This exceeds $b + j$: the difference is at least

$$0.01m - \delta - 1 > 0.0035m + 1 > 0,$$

using $\epsilon = (\delta + 2)/m < 0.0065$.

By the direct low-rank finite CDF lemma and (2),

$$\begin{aligned} \frac{R_n(t)}{m} &< \beta_0 - 3(0.0132) + 0.000008 \\ &= \beta_0 - 0.039592 < \beta. \end{aligned}$$

Therefore $R_n(t) < b$. Every retained odd has profile $\geq t$; no gap is exceptional and every connector list has at least j members. No signature transition budget is needed in this case.

5. Universal common-even-neighbor bound

For every odd pair $u, v \leq n$, write $w = \omega(\text{rad}(uv))$. Since its distinct odd prime factors are at least $3, 5, \dots, 2w + 1$,

$$\begin{aligned} \rho(\text{rad}(uv)) &\geq \prod_{i=1}^w \frac{2i}{2i+1} \\ &\geq \sqrt{\frac{3}{4w+3}}. \end{aligned} \tag{6}$$

The second inequality follows by induction: at step w ,

$$\left[\frac{2w}{2w+1} \right]^2 \geq \frac{4w-1}{4w+3},$$

whose cross-multiplied numerators differ by exactly 1. Also

$$w \leq \frac{2 \log n}{\log 3},$$

so the common even-neighbor count is at least

$$\frac{n-1}{2} \sqrt{\frac{3}{8 \log n / \log 3 + 3}} - \frac{3}{2} \sqrt{n}.$$

This quantity is $> 30\sqrt{n}$ for every $n \geq N$.

For completeness, divide the first term by $\sqrt{n}$ and call it $G(n)$. Its square is a positive constant times

$$\frac{(n-1)^2}{n(8 \log n / \log 3 + 3)},$$

which is increasing for $n \geq 3$: the logarithmic derivative comparison is

$$\frac{n+1}{n-1} > \frac{8/\log 3}{8 \log n / \log 3 + 3}.$$

At N , $\log N < 15$, $\log 3 > 1$, and $\sqrt{N} < 1500$. Thus the square-root factor in $G(N)$ is $> 1/7$, and

$$G(N) > \frac{N-1}{14\sqrt{N}} > \frac{N-1}{21000} > 31.5.$$

Subtracting 1.5 proves the claim. The elementary logarithm inequalities follow from $e > 8/3$ and $e < 3$. This intentionally conservative bound is enough.

6. Small ranks with few missing evens: $x \leq 0.02$, $\beta < 0.1$

If $b + j \leq 30\sqrt{n}$, Section 5 shows that every odd pair has at least $b + j$ common evens. Removing at most b missing evens leaves at least j resources for every gap, so the rank- j condition is automatic.

Otherwise $b + j > 30\sqrt{n}$, hence $\delta < (b + j)/20$. Put $z = \beta + x < 0.12$ and

$$t = \sqrt{1.05z/3}.$$

The unstructured common-even-neighbor bound is strictly greater than

$$3mt^2 - \delta = 1.05(b + j) - \delta > b + j.$$

The direct finite moment bound gives

$$\frac{R_n(t)}{m} < 66.001(1.05z/3)^3 < 2.84z^3. \quad (7)$$

If $\beta \geq x$, then $\beta \geq z/2$ and

$$2.84z^3 < z/2 \leq \beta,$$

since $5.68(0.12)^2 < 1$. Thus no retained low-profile vertices remain.

If $\beta < x$, then $z < 2x$ and (7) gives

$$2 \max\left(0, \frac{R_n(t)}{m} - \beta\right) \leq \frac{2R_n(t)}{m} < 45.44x^3 < x,$$

since $x \leq 0.02$ and $45.44(0.02)^2 < 1$.

In both cases fewer than j gaps have insufficient common-neighbor bounds. There is no signature-transition term because the product-density estimate is unstructured.

7. Conclusion and exact dependency boundary

Sections 3,4,6 cover every feasible b and every rank $1 \leq j \leq m$. The bounds hold for one fixed skeleton order: the adaptive full signature in Section 3, with the same ordering harmless in the unstructured cases. The two prescribed odd endpoints are coprime by the exact Ahlswede–Khachatrian theorem. Ordered Hall supplies every required odd cycle for every $n \geq 2,000,000$.

Part V Exact finite bridge and completion

For $n \leq 5$, the theorem has no target odd cycle length at least 3 and is vacuous. The supplied interval list `interval_cover_6_2300.json` contains 71 adjacent intervals covering every integer from 6 to 2300. Each interval passes the exact criterion in Part I using the first four odd primes. The script `check_interval_cover.py` generates the cover. Independent verification recomputes the common-neighbor profiles by direct gcd counts; a separate C++ verifier recomputes them by exact inclusion–exclusion.

The file `adaptive_bridge_2301_2000000_output.txt` contains 84 adjacent PASS intervals covering every integer from 2301 to 2,000,000. The integer-only program `adaptive_bridge.cpp` verifies the profile-only form of the same interval criterion, using $g_s(z) = \max(z^2, \eta_s(U)z)$ from Part II. It checks all Hall ranks

and all feasible missing-even counts through exact range maxima. No selected subset A is sampled or enumerated.

A separately rebuilt source passes the same 84 intervals. A further independently written verifier uses a linear smallest-prime-factor totient sieve and segment-tree range maxima rather than the original sieve and monotone deque. It also passes all 84 intervals. The separate mathematical audit and checker sources are included in the bundle.

REJECT lines in diagnostic logs refer to wider proposed intervals that failed a sufficient test and were split. Every interval in the final contiguous covers passes. Such rejected proposals are not counterexamples to the theorem and do not leave gaps in the cover.

Combining these exact covers with Part IV proves the stated all- n theorem within this proof candidate. The overlap at $n = 2,000,000$ is deliberate. No extrapolation from finite samples, no unspecified n_0 and no unverified large finite range is used in this combination.

Reproduction and review boundary

Read `README.txt` in the version 3 certificate bundle for exact reproduction commands, source versions, and the certificate inventory. Verify `SHA256SUMS.txt` before running anything, and run the checkers in a fresh copy since some regenerate their outputs. The certificate programs use only standard Python 3 and C++17 facilities (with the compiler's `__int128` extension). No paid service, numerical optimizer, external theorem prover or downloaded executable is required.

Independent AI-assisted checks found no gap in the assembled argument. That does not substitute for external mathematical review or formal verification. The manuscript remains labeled a proof candidate.

Reproduction checklist

From a fresh extraction of the v3 certificate bundle, first verify the manifest, then run:

```
python check_interval_cover.py
python independent_signature_audit.py
python independent_adaptive_density_audit.py
python verify_adaptive_tail_lemma.py
python totient_repair/check_guarded_finite_cdf.py
```

Compile `adaptive_bridge.cpp` using a C++17 compiler with `__int128` support (the supplied command uses `g++ -O3 -std=c++17 -Wall -Wextra -Werror`) and run the resulting program with argument `2000000`. The complete commands for the independent C++ implementations and additional arithmetic and foundation checks are in the bundle's `README.txt`.

Expected results are 71 contiguous passing intervals covering $6 \leq n \leq 2300$, 84 contiguous passing intervals covering $2301 \leq n \leq 2,000,000$, and passing exact rational profile, moment, adaptive-tail and numerical guards. The cases $0 \leq n \leq 5$ are vacuous; Part IV handles every $n \geq 2,000,000$.

References

- [1] R. Ahlswede and L. H. Khachatrian, *Sets of integers and quasi-integers with pairwise common divisor*, Acta Arithmetica 74 (1996), 141–153. DOI: 10.4064/aa-74-2-141-153. Primary paper: <http://matwbn.icm.edu.pl/ksiazki/aa/aa74/aa7424.pdf>. Theorem 2(i), printed p. 149, is the external theorem used here.
- [2] P. Erdős and G. N. Sárközy, *On cycles in the coprime graph of integers*, Electronic Journal of Combinatorics 4(2) (1997), R8. DOI: 10.37236/1323. Primary paper: <https://www.combinatorics.org/ojs/index.php/eljc/article/download/v4i2r8/pdf/>. Its Theorem 1 and following endpoint question concern sufficiently large n . That theorem is historical context, not an input here.
- [3] D. Della Pietra, *Odd cycles at the sharp threshold in the coprime graph*, unrefereed public manuscript, publication commit 1f26276d7850fc9afe68e9d7edecf24aa7cc597d, 27 July 2026. Pinned source: <https://github.com/donalddellapietra/erdos-883-proof/blob/1f26276d7850fc9afe68e9d7edecf24aa7cc597d/paper/main.tex>. Its main theorem has an existential n_0 and assumes $n \geq n_0$. The present proof adapts its architecture and does not claim that method as new.
- [4] Formal Conjectures project, statement formalization of Erdős Problem 883, <https://raw.githubusercontent.com/google-deepmind/formal-conjectures/main/FormalConjectures/ErdosProblems/883.lean>. Part (i) quantifies every n . This statement source is not used as a proof.

Appendix A Review-edition provenance

This PDF is a typeset review edition of the version 3 manuscript dated 4 October 2026 UTC. Its mathematical source is `Erdos883_all_n_proof_candidate_v3.txt`, preserved unchanged in the accompanying review-source archive and in the original certificate bundle. Its SHA-256 is:

`2bd1f1ecf8c1e1802b28642808a003d0b372cdb55bb2dc97964259de0c4ab50e`

The review edition adds the requested author line, an explicit AI-assistance disclosure, bibliographic links, this provenance note, and a reproduction checklist drawn from the existing certificate-bundle README. It replaces plain-text mathematical notation by standard typeset notation and retains the original part-local numbering, including the jump from Section 4 to Section 7 in Part II.

Two diagnostic decimal expansions in Part III are marked as diagnostic approximations rather than exact equalities in version 3; this PDF uses the same diagnostic convention. The exact bounds remain

$$E(n) \leq \frac{972000}{47000000} \quad \text{and} \quad E(n) \leq \frac{3969000}{376000000},$$

respectively. All proof inequalities use these exact rational values. No numerical constant or threshold is replaced by a rounded approximation.

Part II now explicitly sets $p_0 = 2$ when $s = 0$. The 0-signature is empty, $p = p_1 = 3$, and the tail primes are exactly the odd primes. This records the convention already used in the Python and C++ certificate implementations; it changes no bound or algorithm. The original v2 manuscript is preserved byte-for-byte, and the v2-to-v3 textual changes are supplied separately.

The proof candidate's scope is unchanged: the first odd-cycle question, every finite n , and every subset A satisfying the stated strict size bound. It does not assert a new result for the separate complete-tripartite question. External mathematical review and formal verification remain pending, and no novelty, first-solution or priority claim is made.